

Notes on Regression Fundamentals

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1 Regression Fundamentals

1.1 Theorem 3.1.1 - The CEF Decomposition Property

If $Y_i = \mathbb{E}[Y_i|X_i] + \varepsilon_i$, then (i) $\mathbb{E}[\varepsilon_i|X_i] = 0$ which also implies $X_i \perp \mathbb{E}[\varepsilon_i|X_i]$
and (ii) for any function of X_i i.e. $h(X_i)$, $\mathbb{E}[h(X_i)\varepsilon_i] = 0$ which also implies $Cov(h(X_i), \varepsilon_i) = 0$

Proof:

$$\mathbb{E}[\varepsilon_i|X_i] = \mathbb{E}[Y_i - \mathbb{E}[Y_i|X_i]|X_i] = \mathbb{E}[Y_i|X_i] - \mathbb{E}[\mathbb{E}[Y_i|X_i]|X_i] = \mathbb{E}[Y_i|X_i] - \mathbb{E}[Y_i|X_i] = 0$$

$$\mathbb{E}[X_i\mathbb{E}[\varepsilon_i|X_i]] = \mathbb{E}[X_i \cdot 0] = \mathbb{E}[0] = 0 = \mathbb{E}[X_i] \cdot 0 = \mathbb{E}[X_i]\mathbb{E}[0] = \mathbb{E}[X_i]\mathbb{E}[\mathbb{E}[\varepsilon_i|X_i]] \iff X_i \perp \mathbb{E}[\varepsilon_i|X_i]$$

$$\mathbb{E}[h(X_i)\varepsilon_i] = \mathbb{E}[\mathbb{E}[h(X_i)\varepsilon_i|X_i]] = \mathbb{E}[h(X_i)\mathbb{E}[\varepsilon_i|X_i]] = \mathbb{E}[h(X_i) \cdot 0] = \mathbb{E}[0] = 0$$

$$Cov(h(X_i), \varepsilon_i) = \mathbb{E}[h(X_i)\varepsilon_i] - \mathbb{E}[h(X_i)]\mathbb{E}[\varepsilon_i] = 0 - \mathbb{E}[h(X_i)]\mathbb{E}[\mathbb{E}[\varepsilon_i|X_i]] = -\mathbb{E}[h(X_i)]\mathbb{E}[0] = -\mathbb{E}[h(X_i)] \cdot 0 = 0 \quad \text{Q.E.D.}$$

1.2 Theorem 3.1.2 - The CEF Prediction Property

Conditional Expectation Function (CEF), $\mathbb{E}[Y_i|X_i]$, is the Best (in MMSE sense) Predictor of Y_i given X_i .

Proof: Let $m(X_i)$ be any predictor of Y_i given X_i ,

$$\begin{aligned} \arg \min_{m(X_i)} \mathbb{E}[(Y_i - m(X_i))^2] &= \arg \min_{m(X_i)} \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i] + \mathbb{E}[Y_i|X_i] - m(X_i))^2] \\ &= \arg \min_{m(X_i)} \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i]) + (\mathbb{E}[Y_i|X_i] - m(X_i))^2] \\ &= \arg \min_{m(X_i)} \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i])^2 + (\mathbb{E}[Y_i|X_i] - m(X_i))^2 + 2(Y_i - \mathbb{E}[Y_i|X_i])(\mathbb{E}[Y_i|X_i] - m(X_i))] \\ &= \arg \min_{m(X_i)} \{\mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i])^2] + \mathbb{E}[(\mathbb{E}[Y_i|X_i] - m(X_i))^2] + 2\mathbb{E}[\varepsilon_i(\mathbb{E}[Y_i|X_i] - m(X_i))]\} \\ &= \arg \min_{m(X_i)} \{\mathbb{E}[(\mathbb{E}[Y_i|X_i] - m(X_i))^2] + 2\mathbb{E}[\varepsilon_i(\mathbb{E}[Y_i|X_i] - m(X_i))]\} \\ &= \arg \min_{m(X_i)} \{\mathbb{E}[(\mathbb{E}[Y_i|X_i] - m(X_i))^2] + 2 \cdot 0\} \\ &\quad \text{by Theorem 3.1.1 (ii) as } \mathbb{E}[Y_i|X_i] - m(X_i) \text{ is a function of } X_i \\ &= \arg \min_{m(X_i)} \mathbb{E}[(\mathbb{E}[Y_i|X_i] - m(X_i))^2] \\ &= \mathbb{E}[Y_i|X_i] \end{aligned}$$

$$\text{as } \mathbb{E}[Y_i|X_i] - \mathbb{E}[Y_i|X_i] = 0$$

Q.E.D.

$$\begin{aligned}
\frac{\partial(Y_i - X'_i b)^2}{\partial b} \Big|_{b=\beta} &= 0 \\
\frac{\partial(Y_i - X'_i b)^2}{\partial(Y_i - X'_i b)} \frac{\partial(Y_i - X'_i b)}{\partial b} \Big|_{b=\beta} &= 0 \\
2(Y_i - X'_i \beta)^{2-1}(-X_i) &= 0 \\
-2X_i(Y_i - X'_i \beta) &= 0 \\
X_i(Y_i - X'_i \beta) &= 0 \\
\mathbb{E}[X_i(Y_i - X'_i \beta)] &= \mathbb{E}[0] \\
\mathbb{E}[X_i Y_i] - \mathbb{E}[X_i X'_i] \beta &= 0 \\
\beta &= \mathbb{E}[X_i X'_i]^{-1} \mathbb{E}[X_i Y_i]
\end{aligned}$$

So, $\beta = \mathbb{E}[X_i X'_i]^{-1} \mathbb{E}[X_i Y_i] = \arg \min_b \mathbb{E}[(Y_i - X'_i b)^2]$

1.3 Theorem 3.1.3 - The ANOVA Theorem

$$Var(Y_i) = Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[Var(Y_i|X_i)]$$

Proof:

$$\begin{aligned}
Var(Y_i) &= Var(\mathbb{E}[Y_i|X_i] + \varepsilon_i) \\
&= Var(\mathbb{E}[Y_i|X_i]) + Var(\varepsilon_i) + 2Cov(\mathbb{E}[Y_i|X_i], \varepsilon_i) \\
&= Var(\mathbb{E}[Y_i|X_i]) + Var(\varepsilon_i) + 2 \cdot 0 && \text{by Theorem 3.1.1 (ii) as } \mathbb{E}[Y_i|X_i] \text{ is a function of } X_i \\
&= Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[(\varepsilon_i - \mathbb{E}[\varepsilon_i])^2] \\
&= Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[(\varepsilon_i - \mathbb{E}[\varepsilon_i|X_i])^2] \\
&= Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[(\varepsilon_i - \mathbb{E}[0])^2] && \text{by Theorem 3.1.1 (i)} \\
&= Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[(\varepsilon_i - 0)^2] \\
&= Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[\varepsilon_i^2] \\
&= Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[\mathbb{E}(\varepsilon_i^2|X_i)] \\
&= Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[\mathbb{E}((Y_i - \mathbb{E}[Y_i|X_i])^2|X_i)] \\
&= Var(\mathbb{E}[Y_i|X_i]) + \mathbb{E}[Var(Y_i|X_i)] && \text{Q.E.D.}
\end{aligned}$$

1.4 Theorem 3.1.4 - The Linear CEF Theorem (Regression Justification I)

If CEF is a linear function of X_i i.e. $\mathbb{E}(Y_i|X_i) = X'_i b$, then $\mathbb{E}(Y_i|X_i) = X'_i \beta$ where $\beta = \mathbb{E}[X_i X'_i]^{-1} \mathbb{E}[X_i Y_i]$.

Proof: By Theorem 3.1.1 (ii), $\mathbb{E}[X_i \varepsilon_i] = 0$ when $h(X_i) = X_i$. So,

$$\begin{aligned}
\mathbb{E}[X_i(Y_i - \mathbb{E}[Y_i|X_i])] &= 0 \\
\mathbb{E}[X_i(Y_i - X'_i b)] &= 0 \\
\mathbb{E}[X_i Y_i] - \mathbb{E}[X_i X'_i] b &= 0 \\
b &= \mathbb{E}[X_i X'_i]^{-1} \mathbb{E}[X_i Y_i] = \beta
\end{aligned}$$

So, $\mathbb{E}(Y_i|X_i) = X'_i b = X'_i \beta$ where $\beta = \mathbb{E}[X_i X'_i]^{-1} \mathbb{E}[X_i Y_i]$.

Q.E.D.

1.5 Theorem 3.1.5 - The Best Linear Predictor Theorem (Regression Justification II)

$X_i'\beta$ where $\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$ is the Best (in MMSE sense) Linear Predictor of Y_i given X_i .

Proof: By Theorem 3.1.2 (The CEF Prediction Property), $\mathbb{E}[Y_i|X_i]$ is the Best (in MMSE sense) Predictor of Y_i given X_i . In particular, if $\mathbb{E}[Y_i|X_i]$ is a linear function of X_i , Theorem 3.1.4 implies that $\mathbb{E}[Y_i|X_i] = X_i'\beta$ where $\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$. Thus, $\mathbb{E}[Y_i|X_i] = X_i'\beta$ where $\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$ is the Best (in MMSE sense) Linear Predictor of Y_i given X_i . Q.E.D.

1.6 Theorem 3.1.6 - The Regression CEF Theorem (Regression Justification III)

$X_i'\beta$ where $\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$ is the Best (in MMSE sense) Linear Approximation of $\mathbb{E}[Y_i|X_i]$

Proof: From above,

$$\begin{aligned}
 \beta &= \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i] = \arg \min_b \mathbb{E}[(Y_i - X_i' b)^2] \\
 &= \arg \min_b \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i] + \mathbb{E}[Y_i|X_i] - X_i' b)^2] \\
 &= \arg \min_b \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i]) + (\mathbb{E}[Y_i|X_i] - X_i' b)^2] \\
 &= \arg \min_b \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i])^2 + (\mathbb{E}[Y_i|X_i] - X_i' b)^2 + 2(Y_i - \mathbb{E}[Y_i|X_i])(\mathbb{E}[Y_i|X_i] - X_i' b)] \\
 &= \arg \min_b \{\mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i])^2] + \mathbb{E}[(\mathbb{E}[Y_i|X_i] - X_i' b)^2] + 2\mathbb{E}[\varepsilon_i(\mathbb{E}[Y_i|X_i] - X_i' b)]\} \\
 &= \arg \min_b \{\mathbb{E}[(\mathbb{E}[Y_i|X_i] - X_i' b)^2] + 2\mathbb{E}[\varepsilon_i(\mathbb{E}[Y_i|X_i] - X_i' b)]\} \\
 &\quad \text{as } \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i])^2] \text{ is not a function of } b \\
 &= \arg \min_b \{\mathbb{E}[(\mathbb{E}[Y_i|X_i] - X_i' b)^2] + 2 \cdot 0\} \\
 &\quad \text{by Theorem 3.1.1 (ii) as } \mathbb{E}[Y_i|X_i] - X_i' b \text{ is a function of } X_i \\
 &= \arg \min_b \mathbb{E}[(\mathbb{E}[Y_i|X_i] - X_i' b)^2] \quad \text{Q.E.D.}
 \end{aligned}$$

Theorem 3.1.6 implies the OLS estimates of micro-data model is the same as the WLS (weighted by $Pr(X_i = x)$) estimates of grouped-data model.

From the proof of Theorem 3.1.6,

$$\begin{aligned}
 \arg \min_b \mathbb{E}[(Y_i - X_i' b)^2] &= \beta = \arg \min_b \mathbb{E}[(\mathbb{E}[Y_i|X_i] - X_i' b)^2] \\
 \underbrace{\arg \min_b \sum_{i=1}^N (Y_i - X_i' b)^2 \frac{1}{N}}_{OLS} &= \hat{\beta} = \arg \min_b \underbrace{\sum_x (\mathbb{E}[Y_i|X_i = x] - x' b)^2 \overbrace{Pr(X_i = x)}^{\frac{n_x}{N}}}_{WLS}
 \end{aligned}$$

In practice, we always do not know whether the true CEF $\mathbb{E}[Y_i|X_i]$ is a linear or non-linear function of X_i . If we always approximate $\mathbb{E}[Y_i|X_i]$ by $X_i'\beta$ where $\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$. There are two situations:

If $\mathbb{E}[Y_i|X_i]$ is a linear function of X_i , by Theorem 3.1.4 it is equal to $X_i'\beta$ where $\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$. Approximate CEF = true CEF.

If $\mathbb{E}[Y_i|X_i]$ is not a linear function of X_i , approximating $\mathbb{E}[Y_i|X_i]$ with $X_i'\beta$ where $\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$ may not be the best choice (in MMSE sense). Theorem 3.1.6 says $X_i'\beta$ is the best (in MMSE sense) linear approximation of $\mathbb{E}[Y_i|X_i]$ (it holds no matter $\mathbb{E}[Y_i|X_i]$ is linear or not), but it may not be the best approximation of $\mathbb{E}[Y_i|X_i]$. Non-linear approximations (not necessarily the best non-linear approximation) of $\mathbb{E}[Y_i|X_i]$ may be better when $\mathbb{E}[Y_i|X_i]$ is non-linear.

For instance, let $N_k(x)$ be a neighborhood of x having k points, $Ave(Y_i|X_i \in N_k(x)) := \frac{1}{k} \sum_{i: X_i \in N_k(x)} Y_i \rightarrow \mathbb{E}[Y_i|X_i = x]$ when $N, k \rightarrow \infty$ and $k/N \rightarrow 0$ (That is because when $N \rightarrow \infty$, the k points in $N_k(x)$ will be closer to x , which is like $X_i = x$). The Nearest Neighborhood Approximation i.e. $Ave(Y_i|X_i \in N_k(x))$ may be better compared with Best Linear Approximation i.e. $X'_i\beta$ when $\mathbb{E}[Y_i|X_i]$ is not linear.

Moreover, approximating $\mathbb{E}[Y_i|X_i]$ with $X'_i\beta$ when $\mathbb{E}[Y_i|X_i]$ is non-linear will lead to heteroskedasticity, i.e. $Var(\epsilon_i|X_i)$ is a function of X_i . Notice $\epsilon_i = Y_i - X'_i\beta$ may not equal to $\varepsilon_i = Y_i - \mathbb{E}[Y_i|X_i]$.

$$\begin{aligned}
Var(\epsilon_i|X_i) &= \mathbb{E}[(\epsilon_i - \mathbb{E}[\epsilon_i|X_i])^2|X_i] \\
&= \mathbb{E}[(\epsilon_i - 0)^2|X_i] && \text{assume} \\
&= \mathbb{E}[\epsilon_i^2|X_i] \\
&= \mathbb{E}[(Y_i - X'_i\beta)^2|X_i] && X'_i\beta \text{ is the approximate CEF} \\
&= \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i] + \mathbb{E}[Y_i|X_i] - X'_i\beta)^2|X_i] && \mathbb{E}[Y_i|X_i] \text{ is the true CEF} \\
&= \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i]) + (\mathbb{E}[Y_i|X_i] - X'_i\beta)^2|X_i] \\
&= \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i])^2 + (\mathbb{E}[Y_i|X_i] - X'_i\beta)^2 + 2(Y_i - \mathbb{E}[Y_i|X_i])(\mathbb{E}[Y_i|X_i] - X'_i\beta)|X_i] \\
&= \mathbb{E}[(Y_i - \mathbb{E}[Y_i|X_i])^2|X_i] + \mathbb{E}[(\mathbb{E}[Y_i|X_i] - X'_i\beta)^2|X_i] + 2\mathbb{E}[\varepsilon_i(\mathbb{E}[Y_i|X_i] - X'_i\beta)|X_i] \\
&= Var(Y_i|X_i) + \mathbb{E}[(\mathbb{E}[Y_i|X_i] - X'_i\beta)^2|X_i] + 2(\mathbb{E}[Y_i|X_i] - X'_i\beta)\mathbb{E}[\varepsilon_i|X_i] \text{ as } \mathbb{E}[Y_i|X_i] - X'_i\beta \text{ is a constant given } X_i \\
&= Var(Y_i|X_i) + \mathbb{E}[(\mathbb{E}[Y_i|X_i] - X'_i\beta)^2|X_i] + 2(\mathbb{E}[Y_i|X_i] - X'_i\beta) \cdot 0 && \text{by Theorem 3.1.1 (i)} \\
&= Var(Y_i|X_i) + (\mathbb{E}[Y_i|X_i] - X'_i\beta)^2 && \text{as } \mathbb{E}[Y_i|X_i] - X'_i\beta \text{ is a constant given } X_i
\end{aligned}$$

So even $Var(Y_i|X_i)$ is constant over X_i , $Var(\epsilon_i|X_i)$ will change with X_i if the true CEF $\mathbb{E}[Y_i|X_i]$ is not linear (so $\mathbb{E}[Y_i|X_i] \neq X'_i b$ for any b). Hence, $\mathbb{E}[Y_i|X_i] \neq X'_i\beta$ where $\beta = \mathbb{E}[X_i X'_i]^{-1} \mathbb{E}[X_i Y_i]$ but we approximate it as linear. $Var(\epsilon_i|X_i)$ is large at X_i when the gap between $\mathbb{E}[Y_i|X_i]$ and $X'_i\beta$ is large at X_i .

Two ways to handle heteroskedasticity: 1. GLS; 2. robust standard error e.g. White's HC standard error

In this case, GLS is WLS with weight $= Var(\epsilon_i|X_i)^{-1}$.

$$Var(\sigma Var(\epsilon_i|X_i)^{-1/2} \epsilon_i|X_i) = \sigma^2 Var(\epsilon_i|X_i)^{-1} Var(\epsilon_i|X_i) = \sigma^2$$

However, we need to know $Var(\epsilon_i|X_i)$ for WLS, which we usually do not know in practice. So, current advice is to use robust standard error. In special case we may have a good approximation of $Var(\epsilon_i|X_i)$,

$$\begin{aligned}
Y_i &= \mathbb{E}[Y_i|X_i] + \varepsilon_i \\
D_i &= \mathbb{E}[D_i|X_i] + \varepsilon_i && \text{assume } Y_i = D_i
\end{aligned}$$

$$\begin{aligned}
Var(\varepsilon_i|X_i) &= Var(Y_i - \mathbb{E}[Y_i|X_i]|X_i) \\
&= Var(Y_i|X_i) && \text{as } \mathbb{E}[Y_i|X_i] \text{ is a constant given } X_i \\
&= Var(D_i|X_i) \\
&= Pr(D_i = 1|X_i)(1 - Pr(D_i = 1|X_i)) \\
&= \mathbb{E}[D_i|X_i](1 - \mathbb{E}[D_i|X_i])
\end{aligned}$$

As $0 \leq Pr(D_i = 1|X_i = x) \leq 1$, $0 \leq \mathbb{E}[D_i|X_i = x] \leq 1$ for $\forall x$. There are many possible approximations of $\mathbb{E}[D_i|X_i]$. From the most general to the most restrictive: $Ave(D_i|X_i \in N_k(x))$, $g(f(X_i))$, $g(X_i, b)$, $g(X'_i b)$ (first = KNN, second = Generalized

Additive Model, third = Non-linear Regression, fourth = Generalized Linear Model) where $g : \rightarrow [0, 1]$ is called mean function (g^{-1} is called link function). The first two are non-parametric and the remaining are parametric.

$$\begin{aligned} D_i &= g(X_i' b) + \epsilon_i \\ Pr(D_i = 1 | X_i) &= g(X_i' b) \\ Var(\epsilon_i | X_i) &= g(X_i' b)(1 - g(X_i' b)) \end{aligned}$$

If g is an identity function, the model is called Linear Probability Model (LPM). As it is likely that $X_i' b > 1$ or $X_i' b < 0$ for at least one $X_i = x$, g is not likely to be an identity function. If the value of X_i is restricted such that $0 \leq X_i' b \leq 1$, it is still possible that g is an identity function i.e. LPM is still possible.

In LPM case, if true CEF $\mathbb{E}[D_i | X_i]$ is non-linear, but we approximate it with $X_i' b$ (we pick $b = \beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i D_i]$ to MMSE by Theorem 3.1.6). As shown above, this will lead to heteroskedasticity and thus we use WLS with weight = $[X_i' \beta (1 - X_i' \beta)]^{-1}$.

If true CEF $\mathbb{E}[D_i | X_i]$ is linear (e.g. if X_i is saturated), by Theorem 3.1.4 $\mathbb{E}[D_i | X_i] = X_i' \beta$ where $\beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i D_i]$, then approximate CEF = true CEF. However, $Var(\epsilon_i | X_i) = X_i' \beta (1 - X_i' \beta)$. So, heteroskedasticity still exists.

2 Conventional and Robust Standard Error

Assume $\mathbf{X}$ is fixed.

$$\begin{aligned}\Omega_c &= \sigma_\varepsilon^2 (\mathbf{X}'\mathbf{X})^{-1} = \mathbb{E}(\varepsilon_i^2) (\mathbf{X}'\mathbf{X})^{-1} \\ \hat{\Omega}_c &= \hat{\sigma}_\varepsilon^2 (\mathbf{X}'\mathbf{X})^{-1} = \left(\sum_i \frac{\hat{\varepsilon}_i^2}{N} \right) (\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

Let $\Psi := \text{Var}(\varepsilon) = \mathbb{E}(\varepsilon\varepsilon') = \text{diag}(\mathbb{E}(\varepsilon_i^2)) = \text{diag}(\psi_i)$

$$\begin{aligned}\Omega_r &= (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\Psi\mathbf{X}) (\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_i \mathbf{x}_i \mathbf{x}_i' \underbrace{\mathbb{E}(\varepsilon_i^2)}_{\psi_i} \right) (\mathbf{X}'\mathbf{X})^{-1} \\ \hat{\Omega}_r &= (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\hat{\Psi}\mathbf{X}) (\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_i \mathbf{x}_i \mathbf{x}_i' \underbrace{\hat{\varepsilon}_i^2}_{\hat{\psi}_i} \right) (\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

Ω_r is the asymptotic covariance matrix of $\sqrt{N}(\hat{\beta}_{ols} - \beta)$ in linear model (under stochastic $\mathbf{x}_i$ and i.i.d. of $(y_i, \mathbf{x}_i')$).

$$\begin{aligned}\hat{\mathbf{y}} &= \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y} & \mathbf{H} &:= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{ is Hat/Projection matrix} \\ \hat{\varepsilon} &:= \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I}_N - \mathbf{H})\mathbf{y} = \mathbf{M}\mathbf{y} \\ &= \mathbf{M}(\mathbf{X}\beta + \varepsilon) \\ &= \underbrace{\mathbf{M}\mathbf{X}}_{\mathbf{0}}\beta + \mathbf{M}\varepsilon \\ &= \mathbf{M}\varepsilon & \text{where } \mathbf{M} &:= (\mathbf{I}_N - \mathbf{H}) \text{ is the residual marker}\end{aligned}$$

$\mathbf{H}$ and $\mathbf{I}$ are both symmetric

$$\begin{aligned}\mathbf{H} &= \begin{pmatrix} \mathbf{h}'_1 \\ \vdots \\ \mathbf{h}'_N \end{pmatrix} = (\mathbf{h}_1 \cdots \mathbf{h}_N) & \text{the second} &= \text{due to symmetric property} \\ \mathbf{I}_N &= \begin{pmatrix} l'_1 \\ \vdots \\ l'_N \end{pmatrix} = (l_1 \cdots l_N) & \text{the second} &= \text{due to symmetric property} \\ \mathbf{M} &= \mathbf{I}_N - \mathbf{H} = \begin{pmatrix} l'_1 \\ \vdots \\ l'_N \end{pmatrix} - \begin{pmatrix} \mathbf{h}'_1 \\ \vdots \\ \mathbf{h}'_N \end{pmatrix} = (l_1 \cdots l_N) - (\mathbf{h}_1 \cdots \mathbf{h}_N) \\ (\mathbf{m}_1 \cdots \mathbf{m}_N) &= \begin{pmatrix} \mathbf{m}'_1 \\ \vdots \\ \mathbf{m}'_N \end{pmatrix} = \begin{pmatrix} (l_1 - \mathbf{h}_1)' \\ \vdots \\ (l_N - \mathbf{h}_N)' \end{pmatrix} = ((l_1 - \mathbf{h}_1) \cdots (l_N - \mathbf{h}_N)) \\ \hat{\varepsilon} &= \mathbf{M}\varepsilon \\ \begin{pmatrix} \hat{\varepsilon}_1 \\ \vdots \\ \hat{\varepsilon}_N \end{pmatrix} &= \begin{pmatrix} \mathbf{m}'_1 \\ \vdots \\ \mathbf{m}'_N \end{pmatrix} \varepsilon \\ \hat{\varepsilon}_i &= \mathbf{m}'_i \varepsilon\end{aligned}$$

$$\begin{aligned}
\mathbb{E}(\widehat{\varepsilon}_i^2) &= \mathbb{E}(\mathbf{m}_i' \boldsymbol{\varepsilon} \mathbf{m}_i' \boldsymbol{\varepsilon}) \\
&= \mathbb{E}(\mathbf{m}_i' \boldsymbol{\varepsilon} \boldsymbol{\varepsilon}' \mathbf{m}_i) \\
&= \mathbf{m}_i' \mathbb{E}(\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}') \mathbf{m}_i \\
&= \mathbf{m}_i' \boldsymbol{\Psi} \mathbf{m}_i \\
&= (\mathbf{l}_i - \mathbf{h}_i)' \boldsymbol{\Psi} (\mathbf{l}_i - \mathbf{h}_i) \\
&= (\mathbf{l}_i' \boldsymbol{\Psi} - \mathbf{h}_i' \boldsymbol{\Psi}) (\mathbf{l}_i - \mathbf{h}_i) \\
&= \mathbf{l}_i' \boldsymbol{\Psi} \mathbf{l}_i - \mathbf{l}_i' \boldsymbol{\Psi} \mathbf{h}_i - \mathbf{h}_i' \boldsymbol{\Psi} \mathbf{l}_i + \mathbf{h}_i' \boldsymbol{\Psi} \mathbf{h}_i \\
&= \psi_i - \begin{pmatrix} 0 \\ \vdots \\ \psi_i \\ \vdots \\ 0 \end{pmatrix} (\mathbf{h}_i - \mathbf{h}_i') + \mathbf{h}_i' \boldsymbol{\Psi} \mathbf{h}_i \\
&= \psi_i - \psi_i h_{ii} - h_{ii} \psi_i + \mathbf{h}_i' \boldsymbol{\Psi} \mathbf{h}_i \\
&= \psi_i - 2\psi_i h_{ii} + \mathbf{h}_i' \boldsymbol{\Psi} \mathbf{h}_i
\end{aligned}$$

$$\begin{aligned}
\begin{pmatrix} \mathbf{x}_1' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \\ \vdots \\ \mathbf{x}_N' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \end{pmatrix} &= \begin{pmatrix} \mathbf{x}_1' \\ \vdots \\ \mathbf{x}_N' \end{pmatrix} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' = \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' = \mathbf{H} = \begin{pmatrix} \mathbf{h}_1' \\ \vdots \\ \mathbf{h}_N' \end{pmatrix} \\
\mathbf{h}_i' &= \mathbf{x}_i' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \\
\mathbf{h}_i &= \mathbf{h}_i'' = (\mathbf{x}_i' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}')' = \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_i
\end{aligned}$$

$$\mathbf{h}_i' \mathbf{h}_i = \mathbf{x}_i' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_i = \mathbf{x}_i' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_i = h_{ii}$$

h_{ii} is leverage of i obs.

$h_{ii} \in [0, 1]$ and $\sum_{j=1}^K h_{ij} = K$.

If $\boldsymbol{\varepsilon}$ is homoskedastic i.e., $\boldsymbol{\Psi} := \text{Var}(\boldsymbol{\varepsilon}) = \sigma_\varepsilon^2 \mathbf{I}_N \implies \psi_i = \sigma_\varepsilon^2$,

$$\begin{aligned}
\mathbb{E}(\widehat{\varepsilon}_i^2) &= \sigma_\varepsilon^2 - 2\sigma_\varepsilon^2 h_{ii} + \mathbf{h}_i' \sigma_\varepsilon^2 \mathbf{I}_N \mathbf{h}_i \\
&= \sigma_\varepsilon^2 (1 - 2h_{ii} + \mathbf{h}_i' \mathbf{h}_i) \\
&= \sigma_\varepsilon^2 (1 - 2h_{ii} + h_{ii}) \\
&= \sigma_\varepsilon^2 (1 - h_{ii})
\end{aligned}$$

Under homoskedasticity, $\boldsymbol{\Omega}_c$ is correct, and we use $\widehat{\boldsymbol{\Omega}}_c$

$$\begin{aligned}
\mathbb{E}(\widehat{\boldsymbol{\Omega}}_c) &= \left(\sum_i \frac{\mathbb{E}(\widehat{\varepsilon}_i^2)}{N} \right) (\mathbf{X}' \mathbf{X})^{-1} \\
&= \sum_i \frac{\sigma_\varepsilon^2 (1 - h_{ii})}{N} (\mathbf{X}' \mathbf{X})^{-1} \\
&= \sigma_\varepsilon^2 \frac{(\sum_i 1 - \sum_i h_{ii})}{N} (\mathbf{X}' \mathbf{X})^{-1} \\
&= \sigma_\varepsilon^2 \frac{(N - K)}{N} (\mathbf{X}' \mathbf{X})^{-1} \\
&\neq \sigma_\varepsilon^2 (\mathbf{X}' \mathbf{X})^{-1} = \boldsymbol{\Omega}_c
\end{aligned}$$

Thus, $\hat{\Omega}_c$ is biased under homoskedasticity. But it can be adjusted easily.

$$\mathbb{E}(\hat{\Omega}_{c,adj}) = \mathbb{E}\left(\frac{N}{N-K}\hat{\Omega}_c\right) = \frac{N}{N-K}\left(\sum_i \frac{\mathbb{E}(\hat{\varepsilon}_i^2)}{N}\right)(\mathbf{X}'\mathbf{X})^{-1} = \left(\sum_i \frac{\mathbb{E}(\hat{\varepsilon}_i^2)}{N-K}\right)(\mathbf{X}'\mathbf{X})^{-1} = \sigma_\varepsilon^2(\mathbf{X}'\mathbf{X})^{-1} = \Omega_c$$

$\hat{\Omega}_r$ is also biased under homoskedasticity, but it can be adjusted easily by HC2 adjustment (see Choice of $\hat{\psi}_i$ below).

Under homoskedasticity and if there is only one regressor and the model is demeaned i.e., $y_i - \bar{y} = \beta(x_i - \bar{x}) + (\varepsilon_i - \bar{\varepsilon})$

$$\begin{aligned} h_{ii} &= \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i \\ &= (x_i - \bar{x})\left(\sum_i (x_i - \bar{x})^2\right)^{-1}(x_i - \bar{x}) \\ &= \frac{(x_i - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \end{aligned}$$

$$\begin{aligned} \mathbb{E}(\hat{\Omega}_c) &= \sigma_\varepsilon^2 \frac{(N-K)}{N} (\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma_\varepsilon^2 \frac{(N-1)}{N} \left(\sum_i (x_i - \bar{x})^2\right)^{-1} \\ &= \sigma_\varepsilon^2 \left(1 - \frac{1}{N}\right) \frac{1}{Ns_x^2} < \sigma_\varepsilon^2 \quad \text{where } s_x^2 = \sum_i (x_i - \bar{x})^2 / N \text{ and assume } \sum_i (x_i - \bar{x})^2 > 1 \end{aligned}$$

It is clear that when $N \rightarrow \infty$, $\mathbb{E}(\hat{\Omega}_c) = \sigma_\varepsilon^2 / \sum_i (x_i - \bar{x})^2$ i.e., still biased downward/underestimate.

$$\begin{aligned} \mathbb{E}(\hat{\Omega}_r) &= (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_i \mathbf{x}_i \mathbf{x}_i' \sigma_\varepsilon^2 (1 - h_{ii})\right) (\mathbf{X}'\mathbf{X})^{-1} \\ &= \left(\sum_i (x_i - \bar{x})^2\right)^{-1} (\sigma_\varepsilon^2 \sum_i (x_i - \bar{x})^2 (1 - h_{ii})) \left(\sum_i (x_i - \bar{x})^2\right)^{-1} \\ &= \left(\sum_i (x_i - \bar{x})^2\right)^{-1} \sigma_\varepsilon^2 \left(\sum_i (x_i - \bar{x})^2 - \sum_i (x_i - \bar{x})^2 h_{ii}\right) \left(\sum_i (x_i - \bar{x})^2\right)^{-1} \\ &= \left(\sum_i (x_i - \bar{x})^2\right)^{-1} \sigma_\varepsilon^2 \left(1 - \sum_i \frac{(x_i - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} h_{ii}\right) \\ &= \frac{1}{Ns_x^2} \sigma_\varepsilon^2 \left(1 - \sum_i h_{ii}^2\right) = \sigma_\varepsilon^2 \left(1 - \sum_i h_{ii}^2\right) \frac{1}{Ns_x^2} \end{aligned}$$

As $h_{ii} \in (0, 1)$, then $0 < \sum_i h_{ii}^2 < N$ and we assume $\sum_i h_{ii}^2 < 1$. By Jensen's inequality,

$$\begin{aligned} \mathbb{E}(h_{ii}^2) &> (\mathbb{E}(h_{ii}))^2 \\ N^{-1} \sum_i h_{ii}^2 &> (N^{-1} \sum_i h_{ii})^2 = (N^{-1} \sum_i \frac{(x_i - \bar{x})^2}{\sum_i (x_i - \bar{x})^2})^2 = 1/N^2 \\ \sum_i h_{ii}^2 &> 1/N \end{aligned}$$

Thus, $\mathbb{E}(\hat{\Omega}_r) < \mathbb{E}(\hat{\Omega}_c) < \sigma_\varepsilon^2$ i.e., $\mathbb{E}(\hat{\Omega}_c)$ is closer to σ_ε^2 .

2.1 Choice of $\hat{\psi}_i$ for robust standard error

2.1.1 $HC_0 : \hat{\psi}_i = \hat{\varepsilon}_i^2$

$\hat{\Omega}_r = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_i \mathbf{x}_i \mathbf{x}_i' \hat{\varepsilon}_i^2\right) (\mathbf{X}'\mathbf{X})^{-1}$ It is also called Eicker-Huber-White estimator.

$$\begin{aligned} \mathbb{E}(\hat{\Omega}_r) &= (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_i \mathbf{x}_i \mathbf{x}_i' \mathbb{E}(\hat{\varepsilon}_i^2)\right) (\mathbf{X}'\mathbf{X})^{-1} && \text{potential heteroskedasticity} \\ &= (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_i \mathbf{x}_i \mathbf{x}_i' \sigma_\varepsilon^2 (1 - h_{ii})\right) (\mathbf{X}'\mathbf{X})^{-1} \neq \sigma_\varepsilon^2 (\mathbf{X}'\mathbf{X})^{-1} = \Omega_c && \text{If homoskedasticity} \end{aligned}$$

Thus, it is biased in both homoskedasticity and heteroskedasticity case.

$$\mathbf{2.1.2} \quad HC_1 : \hat{\psi}_i = \frac{N}{N-K} \hat{\varepsilon}_i^2$$

$$\hat{\Omega}_{r,HC1adj} = (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \frac{N}{N-K} \hat{\varepsilon}_i^2)(\mathbf{X}'\mathbf{X})^{-1} \text{ The default robust option in Stata.}$$

$$\begin{aligned} \mathbb{E}(\hat{\Omega}_{r,HC1adj}) &= (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \frac{N}{N-K} \mathbb{E}(\hat{\varepsilon}_i^2))(\mathbf{X}'\mathbf{X})^{-1} && \text{potential heteroskedasticity} \\ &= \frac{N}{N-K} (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \sigma_\varepsilon^2 (1 - h_{ii}))(\mathbf{X}'\mathbf{X})^{-1} \neq \sigma_\varepsilon^2 (\mathbf{X}'\mathbf{X})^{-1} = \Omega_c && \text{If homoskedasticity} \end{aligned}$$

Thus, it is biased in both homoskedasticity and heteroskedasticity case.

$$\mathbf{2.1.3} \quad HC_2 : \hat{\psi}_i = \frac{1}{1-h_{ii}} \hat{\varepsilon}_i^2$$

$$\hat{\Omega}_{r,HC2adj} = (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \frac{1}{1-h_{ii}} \hat{\varepsilon}_i^2)(\mathbf{X}'\mathbf{X})^{-1}$$

$$\begin{aligned} \mathbb{E}(\hat{\Omega}_{r,HC2adj}) &= (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \frac{1}{1-h_{ii}} \mathbb{E}(\hat{\varepsilon}_i^2))(\mathbf{X}'\mathbf{X})^{-1} && \text{potential heteroskedasticity} \\ &= (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \frac{1}{1-h_{ii}} \sigma_\varepsilon^2 (1 - h_{ii}))(\mathbf{X}'\mathbf{X})^{-1} && \text{If homoskedasticity} \\ &= (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \sigma_\varepsilon^2)(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma_\varepsilon^2 (\mathbf{X}'\mathbf{X})^{-1} = \Omega_c \\ &= \Omega_r && \text{same under homoskedasticity} \end{aligned}$$

Thus, it is unbiased in homoskedasticity, but biased in heteroskedasticity case.

$$\mathbf{2.1.4} \quad HC_3 : \hat{\psi}_i = \frac{1}{(1-h_{ii})^2} \hat{\varepsilon}_i^2$$

$$\hat{\Omega}_{r,HC3adj} = (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \frac{1}{(1-h_{ii})^2} \hat{\varepsilon}_i^2)(\mathbf{X}'\mathbf{X})^{-1}$$

$$\begin{aligned} \mathbb{E}(\hat{\Omega}_{r,HC3adj}) &= (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \frac{1}{(1-h_{ii})^2} \mathbb{E}(\hat{\varepsilon}_i^2))(\mathbf{X}'\mathbf{X})^{-1} && \text{potential heteroskedasticity} \\ &= (\mathbf{X}'\mathbf{X})^{-1}(\sum_i \mathbf{x}_i \mathbf{x}_i' \frac{\sigma_\varepsilon^2}{1-h_{ii}})(\mathbf{X}'\mathbf{X})^{-1} \neq \sigma_\varepsilon^2 (\mathbf{X}'\mathbf{X})^{-1} = \Omega_c && \text{If homoskedasticity} \end{aligned}$$

Thus, it is biased in both homoskedasticity and heteroskedasticity case.

In the case of heteroskedasticity (i.e., Ω_r is correct), $\hat{\Omega}_c$, $\hat{\Omega}_{c,adj}$, $\hat{\Omega}_r$, $\hat{\Omega}_{r,HC1adj}$, $\hat{\Omega}_{r,HC2adj}$, $\hat{\Omega}_{r,HC3adj}$ are biased in finite sample. However, $\hat{\Omega}_r$, $\hat{\Omega}_{r,HC1adj}$, $\hat{\Omega}_{r,HC2adj}$, $\hat{\Omega}_{r,HC3adj}$ are consistent (to Ω_r) in large sample while $\hat{\Omega}_c$, $\hat{\Omega}_{c,adj}$ are not consistent (to Ω_r) in large sample.

In the case of homoskedasticity (i.e., Ω_c is correct, Ω_r reduces to Ω_c), $\hat{\Omega}_c$, $\hat{\Omega}_r$, $\hat{\Omega}_{r,HC1adj}$, $\hat{\Omega}_{r,HC3adj}$ are biased in finite sample. As shown above, $\hat{\Omega}_{c,adj}$ and $\hat{\Omega}_{r,HC2adj}$ are unbiased in finite sample. $\hat{\Omega}_c$, $\hat{\Omega}_{c,adj}$, $\hat{\Omega}_r$, $\hat{\Omega}_{r,HC1adj}$, $\hat{\Omega}_{r,HC2adj}$, $\hat{\Omega}_{r,HC3adj}$ are consistent (to Ω_c) in large sample.

If we don't know the data generating process, we should always use $\hat{\Omega}_r$, $\hat{\Omega}_{r,HC1adj}$, $\hat{\Omega}_{r,HC2adj}$, $\hat{\Omega}_{r,HC3adj}$ since they are always consistent in large sample. Simulation exercises show that $\hat{\Omega}_r$ performs poorly in finite sample (in both heteroskedasticity and homoskedasticity). On the contrary, $\hat{\Omega}_{r,HC2adj}$ performs well in finite sample (in both heteroskedasticity and homoskedasticity) (see page 306 of MHE).

3 Bootstrap Standard Error

3.1 Reasons to use

3.1.1 If there is no closed form solution for the asymptotic covariance matrix

The asymptotic covariance matrix of $\sqrt{N}(\hat{\beta}_{ols} - \beta)$ in linear model can be derived analytically and easily (under stochastic $\mathbf{x}_i$ and i.i.d. of $(y_i, \mathbf{x}_i')$).

3.1.2 Asymptotic refinements

Reduce finite sample bias because bootstrapping is based on second-order Taylor approximation of characteristic function while Central Limit Theorem is only based on first-order approximation.